

Experiment :. 01

1. Experiment Title:

Demonstration of Basic JavaScript Program

2. Software/Tools Required:

- HTML5 (Markup Language)
- CSS3 (Inline and External Stylesheets)
- JavaScript ES6+ (Form Handling and DOM Manipulation)
- VS Code or any text editor
- Live Server Extension (for local development)
- Image Editor (for profile images)

3. Experiment Program Code:

Index.html

```
<!DOCTYPE html>
<html>
<head>

<title>CSS Types</title>

<link rel="stylesheet" href="style.css">

<style>

/* ===== BODY DESIGN ===== */

body{

    margin:0;

    font-family:'Segoe UI', Arial, sans-serif;

    /* Add your background image here */
    background-image:url("COLLEGE.jpg");

    background-size:cover;

    background-position:center;

    background-repeat:no-repeat;

    min-height:100vh;
```

```
}

/* Dark red overlay */

body::before{

    content:"";

    position:fixed;

    top:0;

    left:0;

    width:100%;

    height:100%;

    background:rgba(120,0,0,0.45);

    z-index:-1;

}

/* ===== FORM CARD ===== */

.container{

    width:420px;

    margin:70px auto;

    background:rgba(255,255,255,0.95);

    padding:35px;

    border-radius:18px;

    box-shadow:0 10px 30px rgba(0,0,0,0.35);
```

```
border-top:8px solid #b71c1c;

}

/* College Heading */

h1{

    text-align:center;

    color:#b71c1c;

    font-size:28px;

    text-transform:uppercase;

    letter-spacing:1px;

    margin-bottom:30px;

}

/* Labels */

label{

    font-weight:600;

    color:#7f0000;

    font-size:16px;

}

/* Input Fields */

input{
```

```
width:100%;

padding:13px;

margin-top:8px;

margin-bottom:20px;

border:2px solid #e0a0a0;

border-radius:8px;

font-size:15px;

box-sizing:border-box;

transition:0.3s;
}

input:focus{

    outline:none;

    border-color:#b71c1c;

    box-shadow:0 0 8px rgba(183,28,28,0.4);
}

/* Button */

button{

    width:100%;

    padding:14px;

    background:linear-gradient(45deg,#b71c1c,#f44336);
```

```
        color:white;

        border:none;

        border-radius:8px;

        font-size:17px;

        font-weight:bold;

        cursor:pointer;

        transition:0.3s;
    }

    button:hover{

        transform:scale(1.04);

        background:linear-gradient(45deg,#7f0000,#d32f2f);
    }

</style>

</head>

<body>

<script>

function give_information(){

let name = document.getElementById("name").value;

let prn = document.getElementById("prn").value;
```

```
if(prn==" " || name==""){  
  
    alert("Please enter Name and PRN.");  
  
    return;  
  
}
```

```
let newWindow=window.open("");
```

```
newWindow.document.write(`
```

```
<html>
```

```
<head>
```

```
<title>Student Document</title>
```

```
<style>
```

```
body{
```

```
background:#fff0f0;
```

```
font-family:'Segoe UI',Arial;
```

```
padding:40px;
```

```
}
```

```
.profile1{
```

```
width:120px;
```

```
height:120px;
```

```

        border-radius:50%;
        object-fit:cover;
        border:4px solid #b71c1c;
        position:absolute;
        top:30px;
        left:30px;
        box-shadow:0 4px 10px rgba(0,0,0,0.3);
    }
    .profile2{
        width:120px;
        height:120px;
        border-radius:50%;
        object-fit:cover;
        border:4px solid #b71c1c;
        position:absolute;
        top:30px;
        right:30px;
        box-shadow:0 4px 10px rgba(0,0,0,0.3);
    }
    .paper{
        width:750px;
        margin:auto;
        background:white;
        padding:40px;
        border-radius:15px;
        box-shadow:0 10px 25px gray;
        border-top:8px solid #b71c1c;

        /* Add these two lines */
        position:relative;
        padding-top:140px;
    }

    h1{

text-align:center;

color:#b71c1c;

}

```

```
h3{  
  
color:#c62828;  
  
border-bottom:2px solid #ffcdd2;  
  
padding-bottom:5px;  
  
}
```

```
p{  
  
font-size:18px;  
  
line-height:1.7;  
  
}
```

```
table{  
  
width:100%;  
  
border-collapse:collapse;  
  
margin-top:20px;  
  
}
```

```
table,th,td{  
  
border:1px solid #999;  
  
}
```

```
th{

background:#b71c1c;

color:white;

}
```

```
th,td{

padding:12px;

font-size:17px;

}
```

```
.footer{

text-align:center;

margin-top:40px;

color:#777;

font-style:italic;

}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="paper">


<h1>

Symbiosis Institute of Technology

</h1>

<h3>Student Information</h3>

<table>

<tr>

<th>Name</th>

<td>${name}</td>

</tr>

<tr>

<th>PRN Number</th>

<td>${prn}</td>

</tr>

</table>

<h3>About Me</h3>
```

<p>
Hello! My name is \${name}, and I am a Third Year B.Tech
student at
Symbiosis Institute of Technology. I am passionate about
technology and
continuously learning new concepts in web development and
programming.
</p>

<p>
This webpage has been created as part of my Web Technology
assignment to
demonstrate the use of HTML, CSS, and JavaScript. It showcases
basic web
design, form handling, and dynamic content generation using
JavaScript.
</p>

<p>
I enjoy exploring new technologies, improving my coding skills,
and building
simple yet attractive web applications as part of my academic
learning.
</p>

<h3>Skills</h3>

- HTML5 & Basic Web Page Development
- CSS3 (Internal & External Styling)
- Basic JavaScript Programming
- C Programming
- Python Programming (Beginner)
- Problem Solving & Logical Thinking
- Microsoft Word, PowerPoint & Excel

<h3>Academic Information</h3>

<table>

```
<tr>
    <th>Course</th>
    <td>B.Tech</td>
</tr>

<tr>
    <th>Year</th>
    <td>Third Year</td>
</tr>

<tr>
    <th>Institute</th>
    <td>Symbiosis Institute of Technology</td>
</tr>

<tr>
    <th>Department</th>
    <td>Computer Science & Engineering</td>
</tr>

<tr>
    <th>Area of Interest</th>
    <td>Web Development, Programming & AI</td>
</tr>
</table>
```

```
<div class="footer">
```

```
Thank You<br>
```

```
Symbiosis Institute of Technology
```

```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

```
`);
```

```
newWindow.document.close();
```

```
}
```

```
</script>
```

```
<div class="container">
```

```
<h1>
```

```
Symbiosis Institute of Technology
```

```
</h1>
```

```
<label>
```

```
Student Name
```

```
</label>
```

```
<input type="text" id="name" placeholder="Enter your name">
```

```
<label>
```

```
PRN Number
```

```

</label>

<input type="text" id="prn" placeholder="Enter PRN">

<button onclick="give_information()">

Submit

</button>

</div>

</body>

</html>

```

Script. Js

```

document.addEventListener('DOMContentLoaded', () => {
    const form = document.getElementById('studentForm');
    const detailsCard = document.getElementById('detailsCard');
    const displayName =
document.getElementById('displayNewName');
    const displayPRN = document.getElementById('displayNewPRN');
    const displaySection =
document.getElementById('displayNewSection');

    form.addEventListener('submit', (event) => {
        // Prevent page reload on form submission
        event.preventDefault();

        // Retrieve values from input fields
        const name =
document.getElementById('studentName').value.trim();

```

```

        const prn =
document.getElementById('studentPRN').value.trim();
        const section =
document.getElementById('studentSection').value.trim();

        // Simple validation check
        if (!name || !prn || !section) {
            alert('Please fill out all fields.');
```

return;

```

        }

        // Showcase student details by updating text content
        displayName.textContent = name;
        displayPRN.textContent = prn;
        displaySection.textContent = section;

        // Reveal the details card with animation
        detailsCard.classList.remove('hidden');
```

// Re-trigger animation if card was already visible

```

        detailsCard.style.animation = 'none';
        detailsCard.offsetHeight; /* trigger reflow */
        detailsCard.style.animation = null;
    });
});
```

Style.css

```

/* Modern variables for colors and animations */
:root {
    --bg-gradient: linear-gradient(135deg, #0f172a 0%, #1e1b4b
100%);
    --card-bg: rgba(30, 27, 75, 0.4);
    --card-border: rgba(255, 255, 255, 0.08);
    --text-primary: #f8fafc;
    --text-secondary: #94a3b8;
    --primary-color: #6366f1;
    --primary-hover: #4f46e5;
    --accent-color: #a855f7;
    --success-color: #10b981;
    --input-bg: rgba(15, 23, 42, 0.6);
    --input-border: rgba(255, 255, 255, 0.1);

```

```
--input-focus: #818cf8;
--shadow-premium: 0 20px 40px -15px rgba(0, 0, 0, 0.5);
--glow-effect: 0 0 20px rgba(99, 102, 241, 0.15);
}

/* Reset and base styles */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: 'Plus Jakarta Sans', sans-serif;
}

body {
  min-height: 100vh;
  background: var(--bg-gradient);
  color: var(--text-primary);
  display: flex;
  justify-content: center;
  align-items: center;
  overflow-x: hidden;
  position: relative;
  padding: 20px;
}

/* Background floating decorations for deep visual depth */
.background-decorations {
  position: absolute;
  width: 100%;
  height: 100%;
  top: 0;
  left: 0;
  pointer-events: none;
  z-index: 1;
  overflow: hidden;
}

.circle {
  position: absolute;
  border-radius: 50%;
  filter: blur(80px);
  opacity: 0.35;
}
```

```

.circle-1 {
  width: 300px;
  height: 300px;
  background: var(--primary-color);
  top: 15%;
  left: 20%;
  animation: float1 15s infinite alternate ease-in-out;
}

.circle-2 {
  width: 350px;
  height: 350px;
  background: var(--accent-color);
  bottom: 15%;
  right: 20%;
  animation: float2 18s infinite alternate ease-in-out;
}

@keyframes float1 {
  0% { transform: translate(0, 0) scale(1); }
  100% { transform: translate(50px, 30px) scale(1.1); }
}

@keyframes float2 {
  0% { transform: translate(0, 0) scale(1); }
  100% { transform: translate(-40px, -50px) scale(1.2); }
}

/* Main Container - Premium Glassmorphism */
.container {
  position: relative;
  z-index: 10;
  width: 100%;
  max-width: 480px;
  background: var(--card-bg);
  backdrop-filter: blur(20px);
  -webkit-backdrop-filter: blur(20px);
  border: 1px solid var(--card-border);
  border-radius: 24px;
  padding: 40px;
  box-shadow: var(--shadow-premium), var(--glow-effect);
}

```

```
        transition: transform 0.4s cubic-bezier(0.16, 1, 0.3, 1),
box-shadow 0.4s ease;
    }

.container:hover {
    transform: translateY(-2px);
    box-shadow: var(--shadow-premium), 0 0 30px rgba(99, 102,
241, 0.25);
}

/* Header style */
header {
    text-align: center;
    margin-bottom: 32px;
}

header h1 {
    font-size: 2rem;
    font-weight: 700;
    background: linear-gradient(135deg, #ffffff 40%, #c7d2fe
100%);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
    letter-spacing: -0.5px;
    margin-bottom: 8px;
}

header .subtitle {
    font-size: 0.9rem;
    color: var(--text-secondary);
}

/* Forms and Inputs */
form {
    display: flex;
    flex-direction: column;
    gap: 20px;
}

.input-group {
    display: flex;
    flex-direction: column;
    gap: 8px;
```

```
        width: 100%;
    }

    .input-row {
        display: flex;
        gap: 16px;
    }

    label {
        font-size: 0.85rem;
        font-weight: 600;
        color: var(--text-secondary);
        text-transform: uppercase;
        letter-spacing: 0.5px;
        transition: color 0.3s ease;
    }

    .input-wrapper {
        position: relative;
        width: 100%;
    }

    input {
        width: 100%;
        padding: 14px 16px;
        background: var(--input-bg);
        border: 1px solid var(--input-border);
        border-radius: 12px;
        color: var(--text-primary);
        font-size: 0.95rem;
        outline: none;
        transition: all 0.3s cubic-bezier(0.16, 1, 0.3, 1);
    }

    input::placeholder {
        color: rgba(148, 163, 184, 0.4);
    }

    /* Custom interactive focus border */
    .focus-border {
        position: absolute;
        bottom: 0;
        left: 50%;
```

```
    width: 0;
    height: 2px;
    background: linear-gradient(90deg, var(--primary-color),
var(--accent-color));
    transition: all 0.4s cubic-bezier(0.16, 1, 0.3, 1);
    border-radius: 2px;
}

input:focus {
    border-color: transparent;
    background: rgba(15, 23, 42, 0.8);
    box-shadow: 0 4px 20px rgba(0, 0, 0, 0.2);
}

input:focus + .focus-border {
    width: 100%;
    left: 0;
}

.input-group:focus-within label {
    color: var(--input-focus);
}

/* Submit Button styling */
button[type="submit"] {
    margin-top: 10px;
    padding: 14px 24px;
    background: linear-gradient(135deg, var(--primary-color) 0%,
var(--accent-color) 100%);
    border: none;
    border-radius: 12px;
    color: #ffffff;
    font-size: 1rem;
    font-weight: 600;
    cursor: pointer;
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 8px;
    box-shadow: 0 4px 15px rgba(99, 102, 241, 0.4);
    transition: all 0.3s cubic-bezier(0.16, 1, 0.3, 1);
    position: relative;
    overflow: hidden;
```

```

}

button[type="submit"]::after {
    content: '';
    position: absolute;
    top: 0;
    left: -100%;
    width: 100%;
    height: 100%;
    background: linear-gradient(90deg, transparent, rgba(255,
255, 255, 0.2), transparent);
    transition: 0.5s;
}

button[type="submit"]:hover::after {
    left: 100%;
}

button[type="submit"]:hover {
    transform: translateY(-2px);
    box-shadow: 0 8px 25px rgba(99, 102, 241, 0.5);
    background: linear-gradient(135deg, #4f46e5 0%, #9333ea
100%);
}

button[type="submit"]:active {
    transform: translateY(1px);
    box-shadow: 0 2px 10px rgba(99, 102, 241, 0.4);
}

.btn-icon {
    width: 18px;
    height: 18px;
    transition: transform 0.3s ease;
}

button[type="submit"]:hover .btn-icon {
    transform: translateX(4px);
}

/* Details Card styling (showcased after form submit) */
.details-card {
    margin-top: 32px;
}

```

```
border-top: 1px solid rgba(255, 255, 255, 0.1);
padding-top: 28px;
animation: slideUp 0.6s cubic-bezier(0.16, 1, 0.3, 1)
forwards;
opacity: 0;
transform: translateY(20px);
}

.details-card.hidden {
  display: none;
}

.details-header {
  display: flex;
  align-items: center;
  gap: 12px;
  margin-bottom: 20px;
}

.details-header h3 {
  font-size: 1.15rem;
  font-weight: 600;
  color: var(--text-primary);
}

.badge {
  background: rgba(16, 185, 129, 0.15);
  color: var(--success-color);
  font-size: 0.75rem;
  font-weight: 700;
  padding: 4px 8px;
  border-radius: 20px;
  border: 1px solid rgba(16, 185, 129, 0.2);
  text-transform: uppercase;
  letter-spacing: 0.5px;
}

.details-content {
  display: flex;
  flex-direction: column;
  gap: 12px;
}
```

```
.detail-item {
  background: rgba(255, 255, 255, 0.02);
  border: 1px solid rgba(255, 255, 255, 0.04);
  padding: 12px 16px;
  border-radius: 12px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  transition: all 0.3s ease;
}

.detail-item:hover {
  background: rgba(255, 255, 255, 0.05);
  border-color: rgba(255, 255, 255, 0.08);
  transform: translateX(3px);
}

.detail-item .label {
  font-size: 0.9rem;
  color: var(--text-secondary);
  font-weight: 500;
}

.detail-item .value {
  font-size: 0.95rem;
  color: var(--text-primary);
  font-weight: 600;
}

@keyframes slideUp {
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

/* Responsive adjustment */
@media (max-width: 480px) {
  .container {
    padding: 24px;
    border-radius: 20px;
  }
}
```

```
.input-row {  
    flex-direction: column;  
    gap: 20px;  
}  
}
```

Project Structure:

- index.html - Main form page with inline CSS and JavaScript
- style.css - External CSS file for additional styling and animations
- Images folder - Contains student profile photos (ME.jpg, Doraemon.jpg, COLLEGE.jpg, main-building-1.avif)

Key Features Implemented:

- Student form with name and PRN input fields
- CSS styling using Inline styles and External stylesheet
- JavaScript form validation before submission
- Dynamic HTML document generation using window.open()
- Interactive profile page with embedded images and information
- Gradient buttons and hover effects
- CSS animations and smooth transitions

4. Output:

The screenshot shows a web browser window with the address bar displaying 'File C:/Users/karti/OneDrive/Desktop/AI_ML_Projects/JS_Lab_1/Exp2/index.html'. The main content area features a dark blue background with a subtle circular pattern. A central white card contains the 'Student Details' form. The form has a title 'Student Details' and a subtitle 'Enter your academic credentials below'. It includes three input fields: 'STUDENT NAME' (filled with 'Kartik Ranekar'), 'PRN NUMBER' (filled with '24070521030'), and 'SECTION' (filled with 'A'). A green 'Submit Details →' button is positioned below the inputs. Below the form, a green tab labeled 'SUBMITTED INFO' is active, followed by the 'Profile Overview' section. This section displays the submitted information in a tabular format: Name: Kartik Ranekar, PRN: 24070521030, and Section: A.

SUBMITTED INFO	
Name:	Kartik Ranekar
PRN:	24070521030
Section:	A

Output Description:

The application displays a form-based interface with the following output:

- Student form with college background image and dark overlay
- Input fields for student name and PRN with validation
- Submit button with gradient effect and hover animation
- Generated student profile document with personal information
- Profile images displayed in circular format with borders
- Student skills and academic information in tabular format

5. Case Study Title:

Real-World Application: Building a Student Portfolio Generator

Case Study Overview:

This case study demonstrates how to create an interactive web application that generates personalized student profile documents. The application combines HTML form handling, CSS styling techniques (both inline and external), and JavaScript DOM manipulation to create a dynamic user experience.

6. Case Study Program Code:

```
Index.html
<!DOCTYPE html>
<html>
<head>

<title>CSS Types</title>

<link rel="stylesheet" href="style.css">

<style>

/* ===== BODY DESIGN ===== */

body{

    margin:0;

    font-family:'Segoe UI', Arial, sans-serif;

    /* Add your background image here */
    background-image:url("COLLEGE.jpg");

    background-size:cover;

    background-position:center;

    background-repeat:no-repeat;

    min-height:100vh;

}

/* Dark red overlay */

body::before{

    content:"";

    position:fixed;

    top:0;

    left:0;

    width:100%;

    height:100%;
```

```

        background:rgba(120,0,0,0.45);

        z-index:-1;

    }

/* ===== FORM CARD ===== */

.container{

    width:420px;

    margin:70px auto;

    background:rgba(255,255,255,0.95);

    padding:35px;

    border-radius:18px;

    box-shadow:0 10px 30px rgba(0,0,0,0.35);

    border-top:8px solid #b71c1c;

}

/* College Heading */

h1 {

    text-align:center;

    color:#b71c1c;

    font-size:28px;

    text-transform:uppercase;

    letter-spacing:1px;

    margin-bottom:30px;

}

/* Labels */

```

```
label{

    font-weight:600;

    color:#7f0000;

    font-size:16px;

}

/* Input Fields */

input{

    width:100%;

    padding:13px;

    margin-top:8px;

    margin-bottom:20px;

    border:2px solid #e0a0a0;

    border-radius:8px;

    font-size:15px;

    box-sizing:border-box;

    transition:0.3s;

}

input:focus{

    outline:none;

    border-color:#b71c1c;

    box-shadow:0 0 8px rgba(183,28,28,0.4);

}

/* Button */
```

```
button{  
    width:100%;  
    padding:14px;  
    background:linear-gradient(45deg,#b71c1c,#f44336);  
    color:white;  
    border:none;  
    border-radius:8px;  
    font-size:17px;  
    font-weight:bold;  
    cursor:pointer;  
    transition:0.3s;  
}
```

```
button:hover{  
    transform:scale(1.04);  
    background:linear-gradient(45deg,#7f0000,#d32f2f);  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<script>
```

```
function give_information(){
```

```
    let name = document.getElementById("name").value;
```

```
let prn = document.getElementById("prn").value;
```

```
if(prn=="" || name==""){
```

```
alert("Please enter Name and PRN.");
```

```
return;
```

```
}
```

```
let newWindow=window.open("");
```

```
newWindow.document.write(`
```

```
<html>
```

```
<head>
```

```
<title>Student Document</title>
```

```
<style>
```

```
body{
```

```
background:#fff0f0;
```

```
font-family:'Segoe UI',Arial;
```

```
padding:40px;
```

```
}
```

```
.profile1 {
```

```
width:120px;
```

```
height:120px;
```

```
border-radius:50%;
```

```
object-fit:cover;
```

```
border:4px solid #b71c1c;
```

```
position:absolute;
```

```
top:30px;
```

```

    left:30px;
    box-shadow:0 4px 10px rgba(0,0,0,0.3);
}
.profile2{
    width:120px;
    height:120px;
    border-radius:50%;
    object-fit:cover;
    border:4px solid #b71c1c;
    position:absolute;
    top:30px;
    right:30px;
    box-shadow:0 4px 10px rgba(0,0,0,0.3);
}
.paper{
    width:750px;
    margin:auto;
    background:white;
    padding:40px;
    border-radius:15px;
    box-shadow:0 10px 25px gray;
    border-top:8px solid #b71c1c;

    /* Add these two lines */
    position:relative;
    padding-top:140px;
}

```

```

h1 {

text-align:center;

color:#b71c1c;

}

```

```

h3 {

color:#c62828;

border-bottom:2px solid #ffcdd2;

padding-bottom:5px;

}

```

```
p{  
  
font-size:18px;  
  
line-height:1.7;  
  
}
```

```
table{  
  
width:100%;  
  
border-collapse:collapse;  
  
margin-top:20px;  
  
}
```

```
table,th,td{  
  
border:1px solid #999;  
  
}
```

```
th{  
  
background:#b71c1c;  
  
color:white;  
  
}
```

```
th,td{  
  
padding:12px;  
  
font-size:17px;  
  
}
```

```
.footer{  
  
text-align:center;  
  
margin-top:40px;  
  
color:#777;  
  
font-style:italic;  
  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="paper">  
  
  
<h1>
```

Symbiosis Institute of Technology

```
</h1>
```

```
<h3>Student Information</h3>
```

```
<table>
```

```
<tr>
```

```
<th>Name</th>
```

```
<td>${name}</td>
```

```
</tr>
```

<tr>

<th>PRN Number</th>

<td>\${prn}</td>

</tr>

</table>

<h3>About Me</h3>

<p>

Hello! My name is \${name}, and I am a Third Year B.Tech student at Symbiosis Institute of Technology. I am passionate about technology and continuously learning new concepts in web development and programming.

</p>

<p>

This webpage has been created as part of my Web Technology assignment to demonstrate the use of HTML, CSS, and JavaScript. It showcases basic web design, form handling, and dynamic content generation using JavaScript.

</p>

<p>

I enjoy exploring new technologies, improving my coding skills, and building simple yet attractive web applications as part of my academic learning.

</p>

<h3>Skills</h3>

HTML5 & Basic Web Page Development

CSS3 (Internal & External Styling)

Basic JavaScript Programming

C Programming

Python Programming (Beginner)

Problem Solving & Logical Thinking

Microsoft Word, PowerPoint & Excel

<h3>Academic Information</h3>

```
<table>
<tr>
  <th>Course</th>
  <td>B.Tech</td>
</tr>

<tr>
  <th>Year</th>
  <td>Third Year</td>
</tr>

<tr>
  <th>Institute</th>
  <td>Symbiosis Institute of Technology</td>
</tr>

<tr>
  <th>Department</th>
  <td>Computer Science & Engineering</td>
</tr>

<tr>
  <th>Area of Interest</th>
  <td>Web Development, Programming & AI</td>
</tr>
</table>
```

```
<div class="footer">
```

```
Thank You<br>
```

```
Symbiosis Institute of Technology
```

```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

```
`);
```

```
newWindow.document.close();
```

```
}
```

```
</script>
```

```
<div class="container">
```

```
<h1>
```

Symbiosis Institute of Technology

```
</h1>
```

```
<label>
```

Student Name

```
</label>
```

```
<input type="text" id="name" placeholder="Enter your name">
```

```
<label>
```

PRN Number

```
</label>
```

```
<input type="text" id="prn" placeholder="Enter PRN">
```

```
<button onclick="give_information()">
```

Submit

</button>

</div>

</body>

</html>

Style.css

/* External CSS File */

```
/* Page Background */
body{
    background:#fff0f0;
}
```

```
/* Form Card Enhancement */
.container{

    border-top:8px solid #b71c1c;

    transition:0.3s;

}
```

```
/* Card Hover Effect */
.container:hover{

    transform:translateY(-5px);

    box-shadow:0 15px 35px rgba(183,28,28,0.35);

}
```

```
/* Button Animation */
```

```
button{

    transition:0.3s ease;
```

```
}
```

```
/* Button Hover */
```

```
button:hover{  
    letter-spacing: 1px;  
}
```

```
/* Input Focus Effect */
```

```
input:focus{  
    outline:none;  
    border:2px solid #b71c1c;  
    box-shadow:0 0 8px rgba(183,28,28,0.4);  
}
```

```
/* Smooth appearance */
```

```
input, button{  
    transition:all 0.3s ease;  
}
```

Core Implementation Details:

HTML Structure (index.html):

- Semantic HTML5 form with input fields and submit button
- Form elements: text inputs for name and PRN
- Embedded JavaScript function (give_information) for form handling
- Dynamic HTML generation using template literals (backticks)
- Background images and overlay styling for visual appeal

CSS Implementation (Inline and External):

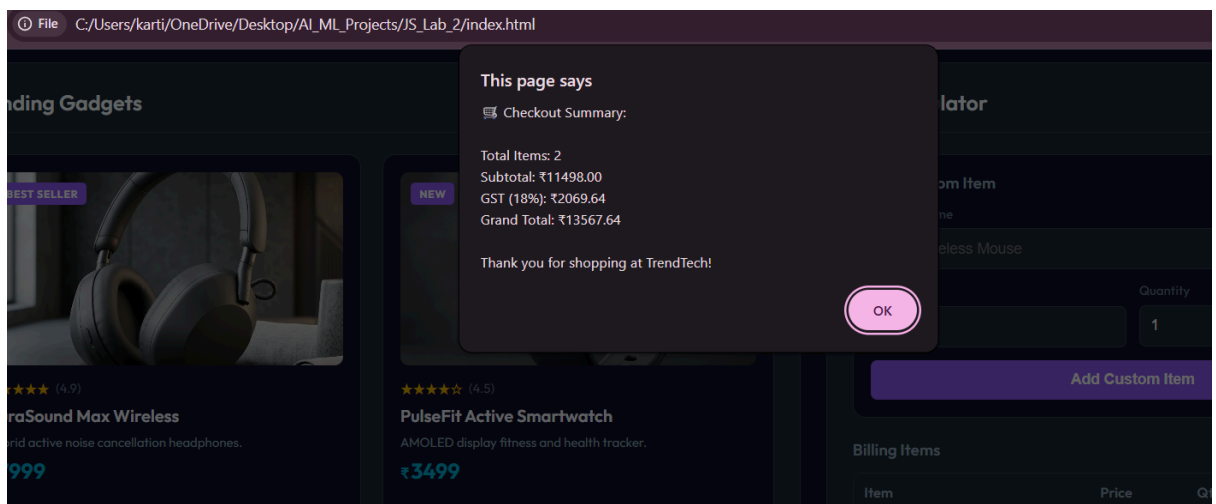
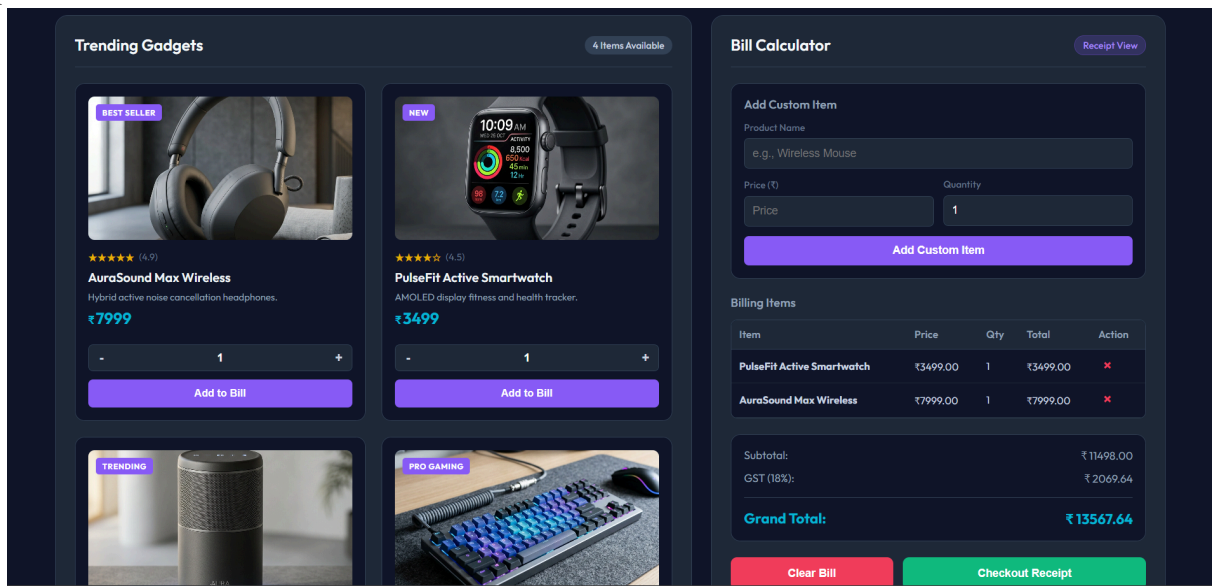
- Inline CSS in <style> tag for form styling:
 - Background image with overlay using ::before pseudo-element
 - Gradient backgrounds for buttons (linear-gradient)
 - Box shadow effects for depth and dimension

- Border-radius for rounded corners
- External CSS file (style.css) for additional enhancements:
 - Hover effects with transform and box-shadow transitions
 - CSS transitions (0.3s ease) for smooth animations
 - Focus states for input fields with border color changes

JavaScript Functionality (script.js):

- Form validation function (give_information):
 - Document.getElementById() for accessing form elements
 - Input validation to check if fields are empty
 - Alert messages for user feedback
- Dynamic content generation:
 - Window.open() to create new browser window
 - Template literals for embedding variables in HTML strings
 - Document.write() to populate new window with HTML content
- Event handling:
 - onclick event handler on submit button

7. Output:



Expected Output Features:

- Initial form page displaying 'Symbiosis Institute of Technology' header
- Form with two input fields (Student Name, PRN Number) and styled buttons
- Form validation alerts if fields are left empty
- Generated student profile document in new window with:
 - Circular profile images (student and decorative images)
 - Student information table with name and PRN
 - Personal information about the student
 - Skills list displaying various technical competencies
 - Academic information table

8. Result/Conclusion:

Results:

The project successfully demonstrates the creation of an interactive web application for generating personalized student profiles. All learning objectives were achieved:

- ✓ Successfully implemented both inline and external CSS styling
- ✓ Demonstrated gradient effects, shadows, and hover animations
- ✓ Implemented form validation and error handling in JavaScript
- ✓ Successfully used `window.open()` and `document.write()` for dynamic content generation
- ✓ Employed template literals for dynamic HTML string creation
- ✓ Demonstrated DOM element access using `document.getElementById()`
- ✓ Created responsive form styling with accessibility considerations

Conclusion:

This Student Profile Generator project successfully demonstrates essential web development techniques combining HTML forms, CSS styling techniques (both inline and external), and JavaScript for dynamic content generation. The project showcases practical skills in:

- Proficiency in CSS styling including gradients, shadows, and animations
- Understanding of CSS types: Inline, Internal, and External stylesheets
- JavaScript form handling and DOM manipulation techniques
- Window object manipulation and dynamic HTML generation
- Input validation and error handling best practices
- Modern JavaScript features like template literals
- User interface design and interactive web application development

The application serves as a foundation for more complex web applications and demonstrates the ability to create production-ready forms with validation, styling, and dynamic content generation using core web technologies.